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11. Output WebGL Content on a Screen

This tutorial shows how to install and modify a container that outputs web browser and webGL content.

Author · Pablo Marquínez · Last revision · 09/25/2024

Overview

The Arduino Portenta X8's processor **NXP® i.MX 8M Mini Processor** is capable of 3D rendering by using OpenGL to process the 3D-related calculations, allowing us to display 3D content on a screen or video output.

In this tutorial, we will render web content from the internet using WebGL and display it on a screen using a USB Hub. We will go through the steps to set up, install, and modify the video output.

Goals

Learn how to setup the Video Output

Learn how to get the required container

Learn how to run the container

Learn how to change the video output

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Required Hardware and Software

Portenta X8

USB-C® cable (USB-C® to USB-A cable)

USB-C® hub with HDMI

External monitor

HDMI cable

Instructions

Install The Container

There are two ways to get the container: through `foundriesFactories` or downloading the container from [portenta-containers repository](#).

With Foundries.io:

If you use [Foundries.io](#), you can switch the current `target` of your device to `x-kiosk-imx8-webgl` by switching the app from a terminal on your computer:

```
1 //Change the app to an e
2 fiocctl devices config up
3
4 //Make a clean installat
5 fiocctl devices config up
6
7 //If you are having issu
8 //You can logout:
9 fiocctl logout
10
11 //Then log in again and
12 fiocctl login
```

You will see the home screen for a

Open the Aquarium 3D from [WebGL samples - Aquarium](#).

With downloaded repository:

If you downloaded the [portenta-containers repository](#), you would need to connect your board directly to your computer and run the `adb shell`, then push the container to your Portenta X8.

Connect to a Wi-Fi®

Check the available Wi-Fi® access points using the `nmcli de wifi` command. You will be able to see an output laying out `BSSID`, `SSID`, and its other elements.

```
1 nmcli de wifi
2
3 //Output
4 IN-USE  BSSID
5         AA:BB:CC:DD:EE:FF
```

The Wi-Fi® details can be saved using the following commands in sequence:

```
1 nmcli c add type wifi co
2 nmcli con modify <custom
3 nmcli con modify <custom
4 nmcli con up <customName
5
6 //To disconnect from a c
7 nmcli con down <customNa
8
9 //To delete a saved conn
10 nmcli c delete <customNa
```

If the LED is illuminating Green, we

connected. If you want to check it in your terminal, you can use the following commands:

```
1 nmcli de
2
3 //Output
4 DEVICE    TYPE        STATE
5 usb0      ethernet    connecte
6 usb1      ethernet    connecte
7 wlan0     wifi        connecte
8 docker0   bridge      connecte
```

The output table will display information regarding active connections and the Wi-Fi® connection we are interested in.

Get Your Board's IP

The IP information of the board can be obtained using `ifconfig wlan0` command. It will show different IP information composed of `inet`, `netmask`, and `broadcastIP`.

```
1 ifconfig wlan0
2
3 //Output
4 wlan0: flags=4163<UP, BROADCAST, MULTICAST>
5         inet <localIP> netmask <netmask> broadcast <broadcastIP>
```

Test your IP connection by exiting the `adb shell`, you can use **CTRL+Z** or type `exit`, then try to connect through **SSH** using the following command:

```
1 ssh -i <key> <username>@<localIP>
```



To connect through SSH, the user's admin password you setup at first access will be requested. If you have trouble connecting with the SSH, please check the troubleshooting section at the end of this tutorial.

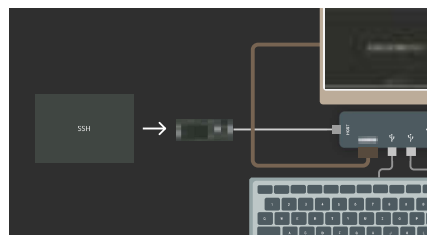
Copy/Push the Docker-Compose.yml

You can push the **docker-compose.yml**, a file that describes the app in ASCII format, from your computer using a terminal on the file's directory. The following command is used to send the composer to the Portenta X8:

```
1 scp <folderName> fio@por
```

Video Output Setup

Now, we need a USB Hub with an available video output connector, such as an HDMI cable. Connect the Portenta X8 to the USB Hub as a Host, the video connector to a display, and the power supply USB to your computer. It is optional, but we could connect a USB mouse to the hub. The setup should look like as follows:





As a reference, a list of validated USB-C® to HDMI hubs that you can use are: [TPX00145](#) and [TPX00146](#).

By default, if you connect the board to a display, you will see the "home screen" with the `Arduino PRO` background wallpaper and a bottom bar with a real-time clock.



You can interact with the interface by plugging USB devices into your hub, like a mouse or a keyboard.



X8 home-screen

Suppose you need to change the resolution of your display/monitor to improve the video output quality. In that case, you need to add a specific resolution to the configuration file of the graphical server (Weston on Portenta X8). To do so, you need to generate the right **Modeline**, i.e., a row that specifies a custom mode for the graphical interface to drive the display correctly.

In the next steps, we provide an example of a 1600 x 758 display running at 60Hz. If you need to modify the modeline and generate a new one, you can use `cvt` command, which is already installed in the Linux image running on your Portenta X8 (see [here](#) to get more information).

To get started in modifying the resolution of your display, connect to your Portenta X8 through ADB (check [this link](#) to learn how to do it).

At this point, you are ready to modify the

`/etc/xdg/weston/weston.ini` file with the `Vim` command as follows:

```
1 sudo vim /etc/xdg/weston/
```

You can now add the following lines to the `weston.ini` file:

```
1 [output]
2 name=DP-1
3 mode=98.00 1600 1680 184
```

Save the file and exit. To see the changes in place, you have to reboot your Portenta X8 using the command `sudo systemctl reboot`. When the board gets started again, you will be able to see your display with the right resolution.

Running The Image

If you obtained the app descriptor, structured through *docker-compose.yml*, from **Foundries.io**, it will run automatically after a few seconds.

On the other hand, if you copied from the repository, you will need to initialize with **docker** by accessing your Portenta X8 through SSH, going

copied it, and running it from that directory using the following commands:

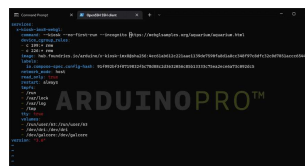
```
1 //Connect to your device
2 ssh fio@<portentaX8-IP>
3
4 //Change directory
5 cd <composerPath>
6
7 //Compose with docker
8 docker compose up --detached
9
10 //Stop the docker-composer
11 docker compose stop
```

Edit The Output

It is possible to change the web output URL by editing the `docker-compose.yml` file using the following commands:

```
1 //Connect to your device
2 ssh fio@<portentaX8-IP>
3
4 //Change directory
5 cd <composerPath>
6
7 //Edit the file with VIM
8 vim docker-compose.yml
```

Once inside the **VIM** editor, you will need to press **insert** and replace the URL as shown in the screenshot to edit the file.



VIM editing docker- compose.yml

Press the **ESC** key to save the changes and type `:wq`. This will write and quit the **VIM** editor. After editing the file, you will need to compose the container again to make the changes take effect.

Conclusion

In this tutorial, we went through how to connect the board and display something on a screen. Using a container from FoundriesFactories or downloading it and uploading it to your Portenta X8. Lastly, we showed how to edit the video output by editing the container.

Next Steps

Make your own HTML content, push it to your device, and output the desired content.

You could make an app that shows information about the weather on the web and have that on a display.

Troubleshooting

If you tried to connect with `ssh` and you get a **fingerprint** issue, you will need to remove the IP and fingerprint on your `.ssh` file. On Windows, the file is located at

```
C:\Users\  
<yourUsername>\.ssh\known  
_hosts
```

and try again with the **ssh**

```
1 $portenta-x8: ssh-keygen
```

Suggest changes

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